Optimizing User Experience: Global Dining Trends Analysis

Food Tastes Best When It's On Time

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Introduction

Background

The company, a global restaurant review aggregator similar to Zomato, is experiencing a decline in user engagement and feedback. To address this, the company seeks to analyze its extensive dataset of restaurant information, including ratings, cuisines, locations, price ranges, and services, to uncover insights that enhance customer experience and support restaurants in improving their offerings.

Problem

The challenge is to identify trends and preferences in dining behavior to:

1.Improve user satisfaction and engagement.

2.Highlight high-quality restaurants.

3.Recommend strategies to introduce new features and services.

Project Scope

This project aims to analyze global restaurant data to uncover actionable insights that can enhance the user experience and support restaurants in improving their services. The analysis will provide a deeper understanding of customer preferences and market trends by examining factors such as cuisines, locations, price ranges, ratings, and services like table bookings and online delivery. The project will identify patterns in dining behavior, highlight high-performing restaurants, and explore the impact of various features on user engagement. Additionally, the findings will inform strategies to introduce personalized recommendations, optimize restaurant visibility, and implement features that cater to user needs, ultimately driving customer satisfaction and platform engagement.

Data Overview

The dataset used for this project is sourced from Zomato and contains extensive information about restaurants from various cities worldwide. The dataset is structured with the following columns, which offer detailed insights into the restaurants' characteristics, performance, and customer engagement. Below is an overview of the key attributes:

1. **Restaurant ID:** A unique identifier for each restaurant.
2. **Restaurant Name:** The name of the restaurant.
3. **Country Code:** A numerical code indicating the country of the restaurant.
4. **City:** The city where the restaurant is located.
5. **Cuisines:** Types of cuisines offered, separated by commas.
6. **Average Cost for Two:** The average cost of a meal for two people in the restaurant's local currency.
7. **Has Table Booking:** Indicates if the restaurant allows table bookings.
8. **Has Online Delivery:** Indicates if the restaurant offers online delivery.
9. **Is Delivering Now:** Reflects whether the restaurant is currently offering delivery services.
10. **Price Range:** A numerical value representing the restaurant's price range (1 to 4).
11. **Aggregate Rating:** The overall rating of the restaurant based on user reviews.
12. **Votes:** The total number of votes the restaurant has received from users.

Methodology

Data sources

1.SQL

2.AWS

3.Data Scraping

4.Local data sources

•Data wrangling

1.Data understanding

2.Data cleaning

3.Data merging and joining

4.Data manipulation

•Data analysis

1.Finding the trends and patterns

•Data visualization

Because food tastes best when it's on time!

Dashboard Overview

The Zomato Dashboard provides a comprehensive analysis of restaurant performance, customer behavior, and market trends, enabling data-driven decision-making. Designed with interactivity and user-friendliness in mind, it highlights critical metrics such as restaurant distribution, customer preferences, cuisine trends, and cost insights. The dashboard is segmented into visually engaging pages, each focusing on specific aspects of Zomato's global operations.

Key Highlights

- **Restaurant Coverage**: Interactive maps and visuals showcasing restaurant density and distribution across countries and cities.
- **Customer Preferences**: Insights into popular cuisines, online delivery, and table booking trends.
- **Performance Metrics**: KPI cards summarizing total restaurants, average ratings, and total votes.
- **Cost Analysis**: Visualizations exploring price range distributions and regional cost trends.
- **Data Interactivity**: Filters for drilling down by country, city, price range, and cuisines for tailored analysis.

Stakeholders

The primary stakeholder for this dashboard is **Zomato Management**, who can leverage it to:

1. **Identify Key Markets**: Determine regions with high restaurant density and customer engagement for expansion and investment opportunities.
2. **Optimize Offerings**: Analyze customer preferences to refine menu offerings, pricing strategies, and service features like online delivery and table booking.
3. **Monitor Performance**: Track high-performing cities, cuisines, and restaurants to replicate success across other markets.
4. **Strategic Decision-Making**: Use data insights to enhance customer satisfaction, boost operational efficiency, and stay ahead of competitors.

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This dashboard serves as a powerful tool for Zomato's leadership team to make informed, strategic decisions that align with the company's growth objectives.

Technical Processes

- •Use pivot tables for summarizing data.
- •Calculate averages, variances, and growth rates.
- •Create charts and graphs for visual representation.
- •Apply filters and sorting for specific analyses.
- •Use functions like Count and Sum IF for data aggregation.

Key Findings

1. Distribution of Ratings

•A majority of restaurants have ratings in the **3–4.5 range**, reflecting a generally positive customer experience.

2. Online Delivery Services

•**26% of restaurants** offer online delivery services.
•Restaurants providing online delivery have an **average rating of 3.2**, compared to **2.5** for those without.
•Online delivery is more prevalent in metropolitan cities, contributing to higher customer engagement.

3. Average Cost for Two Across Cities

•The average cost for two varies significantly by location:

- **Singapore** has the highest average cost (**INR 9,594**).
- Smaller cities, like **Ankara, ÛÁstanbul** and **Miller**, have the lowest costs.

•The global average cost for two is **INR 817**.

4. Top Cuisines

•The five most popular cuisines globally are:

1. **North Indian**
2. **Chinese**
3. **Fast Food**
4. **Bakery**
5. **Cafe**

•Restaurants offering multiple cuisines tend to attract higher ratings.

Key Findings

5. Table Booking Services

•Only **12% of restaurants** offer table booking services.

6. Top Restaurants by Votes

•The most popular restaurants (by votes) are:
- **Barbeque Nation:** 28,142 votes
- **AB's - Absolute Barbecues:** 13,400 votes
- **Toit:** 10,934 votes

•These restaurants also maintain high ratings, indicating strong customer satisfaction.

7. Correlation Between Cost and Ratings

•There is a **No correlation** between the average cost for two and aggregate ratings:
- Higher-priced restaurants generally receive better ratings, likely due to superior service and food quality.

8. Regional Trends

•**Delhi, Gurgaon**, and **Noida** strongly prefer North Indian cuisine.

9. Geographic Spread

•Restaurants are densely clustered in urban centers, with significant geographic variation in pricing, cuisine, and service offerings.
•Popular localities, such as **Inner City** (South Africa) and **Quezon City** (Phillipines), boast high ratings.

Recommended Analysis

1. **Rating Distribution:** Examine aggregate ratings to evaluate overall restaurant performance.
2. **Online Delivery:** Compare restaurants offering and not offering online delivery services.
3. **Average Cost:** Analyze the average cost for two people across cities to identify high-cost and low-cost regions.
4. **Cuisine Preferences:** Determine globally popular cuisines offered by restaurants.
5. **Price Range:** Compare price ranges in localities to assess affordability trends.
6. **Table Booking Effect:** Analyze how table booking availability influences ratings and compare average ratings for restaurants with and without this feature.
7. **Top-Rated Restaurants:** Identify and rank top restaurants based on votes and ratings for promotional purposes.
8. **City-Level Analysis:** Assess geographical diversity by counting cities and mapping restaurant distribution.
9. **Cuisines & Cost:** Investigate the correlation between the variety of cuisines and the average cost for two people.
10. **Online Delivery by Locality:** Analyze online delivery adoption rates across localities to identify trends.
11. **Price vs. Ratings:** Explore how price range correlates with aggregate ratings using visualizations.
12. **Geospatial Analysis:** Map restaurant locations to identify clusters, underserved areas, and expansion opportunities.

Conclusion

The analysis of global dining trends using Zomato's restaurant dataset provides valuable insights into customer preferences, regional variations, and service offerings. Key findings reveal that higher-rated restaurants often offer premium services like table booking and online delivery, with North Indian and Chinese cuisines being the most popular globally. Urban centers dominate the market with diverse culinary options and higher customer engagement, while smaller cities showcase affordability and niche preferences. The positive correlation between cost and ratings highlights that customers associate quality with pricing, while the geographic spread of restaurants underscores the importance of hyper-local strategies. These insights suggest actionable opportunities for enhancing the platform, such as promoting high-rated restaurants, expanding online delivery in underserved areas, and introducing personalized recommendations to improve user engagement and satisfaction. This data-driven approach positions the company to strengthen its competitive edge and deliver a superior dining experience for its users.

Thank you

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